Journal of Veterinary Internal Medicine

Journal of Veterinary Internal Medicine Open Access ACVIM American College of Veterinary Internal Medicine

STANDARD ARTICLE OPEN ACCESS

Small Animal Internal Medicine Immunology

Evaluation of Immune System Components in Dogs With Protein-Losing Enteropathy Compared to Healthy Controls

Emily Moore | Kyan Thelen Strong | Sara A. Jablonski

Department of Small Animal Clinical Sciences, College of Veterinary Medicine, Michigan State University, East Lansing, Michigan, USA

Correspondence: Sara A. Jablonski (sjw@msu.edu)

Received: 13 May 2025 | **Revised:** 5 September 2025 | **Accepted:** 5 September 2025

Funding: This work was supported by the Equipment Grants Program, award #2022-70410-38419, from the U.S. Department of Agriculture (USDA) and National Institute of Food and Agriculture (NIFA).

Keywords: canine | immunology | lymphocyte | protein-losing enteropathy

ABSTRACT

Background: Immune system abnormalities including hypogammaglobulinemia and T-cell deficiency occur in humans with protein-losing enteropathy (PLE). It is unknown whether similar abnormalities occur in dogs with PLE.

Objective: To evaluate serum immunoglobulin (Ig) concentrations and immune cell populations in dogs with PLE (with histologic evidence of chronic inflammatory enteropathy, intestinal lymphangiectasia (IL), or both) compared to healthy controls (HC).

Animals: Eighteen dogs with PLE and 18 HC dogs.

Methods: Prospective study. Serum IgA, IgG, and IgM concentrations were measured via ELISA in treatment-naïve dogs with PLE and compared to concentrations in HC dogs. RNA gene expression of specific immune cell surface markers in peripheral blood mononuclear cells (PBMCs) was measured in both groups by quantitative PCR.

Results: Dogs with PLE had lower concentrations of serum IgG compared to HC dogs (4.5 mg/mL, range 0.67–22.4 mg/mL vs. 19 mg/mL, range 1.8–80.3 mg/mL; $p < 0.001$). Serum IgM concentrations were also lower in dogs with PLE versus HC (2.4 mg/mL, range 0.0009–53.1 mg/mL vs. 14.2 mg/mL, range 2.1–172.8 mg/mL; $p = 0.002$). Expression of CD3e (0.24, range 0.003–1.1 vs. 0.92, range 0.41–3.2; $p < 0.001$), CD5 (0.17, range 0.01–0.46 vs. 0.94, range 0.23–5; $p < 0.001$), and CD8 (0.47, range 0.06–1.7 vs. 0.92, range 0.32–2.4; $p = 0.007$) were reduced in dogs with PLE compared to their mean absolute expression in HC dogs.

Conclusions and Clinical Importance: Dogs with PLE have quantitative reductions in immune system components, similar to humans with IL. These abnormalities in immune system components might be considered in the management and monitoring of dogs with PLE.

1 | Introduction

Protein-losing enteropathy (PLE) in dogs is a syndrome of uncompensated protein loss from the vascular pool due to a diseased or dysfunctional intestinal mucosa [1, 2]. Although multiple disorders could result in canine PLE syndrome, chronic non-neoplastic small intestinal disease is the predominant cause and is characterized histologically by varied inflammatory and morphologic lesions, including intestinal lymphangiectasia (IL) [2]. Intestinal lymphangiectasia often

Abbreviations: 25OHD, 25-hydroxyvitamin-D; BSA, bovine serum albumin; CCECAI, canine chronic enteropathy activity index; CIE, chronic inflammatory enteropathy; GI, gastrointestinal; IBD, inflammatory bowel disease; Ig, immunoglobulin; IL, intestinal lymphangiectasia; MSU-VMC, Michigan State University Veterinary Medical Center; PBS, phosphate buffered saline; PIL, primary intestinal lymphangiectasia; PLE, protein-losing enteropathy; PMBCs, peripheral blood mononuclear cells; qPCR, quantitative PCR; RI, reference interval.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Journal of Veterinary Internal Medicine* published by Wiley Periodicals LLC on behalf of American College of Veterinary Internal Medicine.

occurs along with intestinal inflammation and is identified in approximately 50% of all cases of PLE described in the literature [2], though several studies suggest the actual incidence is considerably higher [3, 4].

Intestinal lymphangiectasia in humans occurs primarily (Waldmann's disease) or secondary to various conditions including inflammatory bowel disease (IBD), portal hypertension, and Fontan surgery complications [5–15]. A variety of immunologic deficits including T-cell and B-cell depletion, decreases in immunoglobulins, impaired or altered cytokine responses, and decreased proliferative responses to mitogens are described in humans with both PIL and secondary IL [15–25]. Proposed theories for these immunodeficiencies include enteric loss of immunoglobulins and lymphocytes, disrupted lymphocyte trafficking, and activation of residual T-cells leading to T-cell death with lack of thymic compensation [18–20, 23, 25].

In general, the frequency of opportunistic infections in humans with IL and confirmed or suspected immunodeficiency appears to be low [20, 23, 26]. However, individual cases of severe infections have been documented [27–29], including one report of opportunistic infections in seven children with primary IL [30]. Of note, the treatment of IL in humans is predominately based on dietary therapy, and broad immunosuppression is generally not utilized [31–34]. In the last decade or so, evidence has been mounting that dietary therapy is critical to the management of canine PLE [35–39], and aggressive immunosuppression might not improve outcomes [40]. Despite this, many dogs with PLE are still treated with broad immunosuppressive therapy, in some instances regardless of the underlying etiology, or if the etiology is unknown [2]. If some dogs with PLE and IL have similar immunologic changes to those observed in humans with IL, careful monitoring for infection could be warranted, and these abnormalities could be considered in disease management, particularly when weighing the risks and benefits of immunosuppressive treatment.

The purpose of this study was to measure serum immunoglobulin (Ig) concentrations and evaluate immune cell populations in a group of dogs with PLE compared to age- and weight-matched healthy control (HC) dogs. Our hypothesis was that Ig concentrations, as well as CD4+ and CD8+ T-lymphocyte expression, would be decreased in a cohort of dogs with PLE when compared to HC dogs.

2 | Material and Methods

2.1 | Sample Size

A sample size calculation was performed using the primary endpoints of serum IgM and IgG concentrations. Data from previous studies was used to assist in the sample size calculations [19, 22]. For IgG, a conjectured difference of 6 mg/mL between dogs with PLE and HC dogs was used, with a common standard deviation of 2 mg/mL. For IgM, a conjectured difference of 1.4 mg/mL between dogs with PLE and HC dogs was used, with a common standard deviation of 0.5 mg/mL. For all endpoints, sample size calculation was performed for a two-sided *t*-test with alpha fixed at 0.05. With a desired power of 0.9, a sample size of $n = 8$ and $n = 10$ was sufficient. We elected to enroll 18 dogs per group to address possible sample quality or processing issues.

2.2 | Dogs

This was a prospective study. Dogs presented to Michigan State University Veterinary Medical Center (MSU-VMC) for evaluation of PLE syndrome were eligible for inclusion. Dogs were presented to the Small Animal Internal Medicine at MSU-VMC either on an outpatient referral basis or as an inpatient transfer. Dogs were required to have clinical signs of gastrointestinal disease (e.g., vomiting, diarrhea, weight loss, and hyporexia) for at least 3 weeks, hypoalbuminemia (serum albumin concentration < 2.5 g/dL; reference interval [RI], 2.8–3.6 g/dL), and a canine chronic enteropathy clinical activity index (CCECAI) score > 3 [41]. Intestinal parasitism was excluded with empiric anthelmintic treatment, fecal flotation via zinc centrifugation, or both. A baseline cortisol concentration > 55 nmol/L (2 μg/dL) or post ACTH-stimulated cortisol concentrations > 138 nmol/L (5 μg/dL) was required to exclude hypoadrenocorticism [42]. In addition, dogs were required to have serum bile acid concentrations within the reference interval, normal trypsin-like immunoreactivity concentrations, and either a negative urine dipstick for protein detection or a urine protein to creatinine ratio < 0.5 [43]. Non-GI causes of clinical signs and hypoalbuminemia were also excluded with routine hematologic and serum biochemical assessments and an abdominal ultrasound examination. Because of the amount of blood required to be collected, dogs weighing < 3 kg were excluded. All PLE dogs meeting these criteria underwent gastrointestinal endoscopic examination with collection of duodenal and, whenever possible, ileal mucosal biopsies. Histopathologic assessments were required to have no evidence of an infectious or neoplastic enteropathy. All histologic evaluations were performed by a single board-certified veterinary pathologist, and evaluations were performed using a standardized scoring system [44]. The type and degree of inflammatory infiltrate were recorded. The degree of lacteal dilation was also recorded. If the lacteal occupied 0%–25% of the villus width, the degree was mild, 25%–50% was moderate, and 50%–75% was severe. Scores for lacteal dilation were based on the most severely affected villus in each case [45].

After recruitment of all PLE dogs, age-, sex-, and weight-matched healthy control dogs were recruited from staff and student-owned dogs at the MSU-VMC. Control dogs were required to be clinically healthy without evidence of chronic gastrointestinal disease within the previous 6 months and with a current CCECAI score ≤ 3, an unremarkable physical examination, and no important abnormalities detected on routine hematology and serum biochemical profile. Control dogs were excluded if they had historic gastrointestinal disease or CCECAI > 3. Both PLE and control dogs were excluded if they had received exogenous glucocorticoids or other immunosuppressive medications (e.g., oclacitinib, cyclosporine, chlorambucil, mycophenolate, azathioprine, leflunomide) within the previous 28 days.

2.3 | Experimental Protocol

The following experimental protocol was approved by the Institutional Care and Use Committee at the MSU-VMC (IACUC ID #PROTO202200456). Written consent was provided by each owner (for both PLE and HC dogs) before study enrollment.

After enrollment, an owner questionnaire (Supporting Information, File S1) was utilized to aid in calculating the CCECAI score for both PLE and HC dogs, and the score was recorded. Purina fecal score [46], body condition score (BCS) [47], and muscle condition score (MCS) [48] were also recorded. A total of 10 mL of whole blood was collected via jugular venipuncture from all dogs for measurement of serum immunoglobulin concentrations and for mononuclear cell characterization. Two milliliters of blood were placed into a sterile red top tube, and the additional 8 mL of blood was placed into a BD Vacutainer CPT mononuclear cell preparation tube (Becton, Dickinson and Company, Franklin Lakes, NJ). Serum from the sterile red top tube was separated and stored at −80°C for future batch analysis. The CPT tubes were processed immediately following collection.

2.4 | PBMC Isolation

Peripheral blood mononuclear cells (PBMC) were isolated via CPT tubes (BD Biosciences, New Jersey, USA) following manufacturer protocol. Cells were counted using Trypan blue dilution and an automated cell counter (Corning, Arizona, USA). Cells were stored under liquid nitrogen in cryopreservation medium containing dimethylsulfoxide (DMSO) and fetal bovine serum (FBS) for flow cytometry analysis and in RNA*later* (Thermo Fisher Scientific, Massachusetts, USA) for gene expression analysis.

2.5 | Assays

2.5.1 | Immunoglobulin Measurement

Serum immunoglobulin expression was quantified using commercially available kits according to the manufacturer's instructions (Canine IgA ELISA Kit [Abcam, Massachusetts, USA], Canine IgG ELISA Kit [Abcam, Massachusetts, USA], and Canine IgM ELISA Kit [Antibodies.com, Missouri, USA]).

2.5.2 | Surface Marker Expression Analysis

Cryopreserved cells were thawed following standard protocol and washed once with phosphate buffered saline (PBS) containing 7.55 μM bovine serum albumin (BSA) and 5 mM ethylenediaminetetraacetic acid (EDTA). Viable cells were identified with a LIVE/DEAD Fixable Yellow Dead Cell Stain (Thermo Fisher Scientific, Massachusetts, USA) and then blocked with an Fc-receptor inhibitor (Anti-Canine Fc Receptor Block, Thermo Fisher Scientific, Massachusetts, USA). At this time, it was determined that the cell numbers in the dogs with PLE were insufficient for accurate analysis, and the experiment was halted.

2.5.3 | Gene Expression Analysis

Total RNA was extracted from PBMCs using the RNeasy Mini Kit (Qiagen, Hilden, Germany) according to the manufacturer's instructions. cDNA was synthesized with the same quantity of total RNA per sample using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Massachusetts, USA) according to the manufacturer's instructions. Quantitative PCR (qPCR) was performed using the *Power* SYBR Green Master Mix (Applied Biosystems, Massachusetts, USA), with samples run in triplicate on the QuantStudio7 Real-Time PCR machine (Applied Biosystems, Massachusetts, USA). The following cycling conditions were used: 50°C for 2 min, 95°C for 10 min, followed by 40 cycles of 95°C for 15 s and 60°C for 60 s. *B2M* and *GAPDH* were used as housekeeping gene controls. Data are presented as fold change relative gene expression ($2^{-\Delta\Delta CT}$) of CD3e, CD4, CD5, CD8, and CD21 compared to the healthy control group. Gene primer sequences are listed in Table S1. A pool of cDNA from healthy control dogs was run on each plate for B2M expression to determine the intra- and inter-assay coefficient of variation (Figure S1).

2.6 | Statistical Analyses

Data were assessed for normality with boxplot analyses and Shapiro–Wilk testing and reported as median and range, or mean ± SD, where applicable. Age, weight, and relevant baseline biochemical data were compared between PLE dogs and HC with the Mann–Whitney *U* test. Serum immunoglobulin (IgA, IgG, and IgM) concentrations were compared between PLE dogs and HC dogs with Mann–Whitney *U* tests. The relationships between serum immunoglobulin (IgA, IgG, and IgM) concentrations and age and weight were assessed in both PLE and HC dogs via Spearman's correlation. The total number of mononuclear cells obtained from CPT tube collection was compared between PLE dogs and HC dogs with a Student's *t*-test. Assessment of immune cell populations via surface marker expression analysis (flow cytometric analysis) was unable to be performed because of an insufficient number of viable cells available for flow cytometric analysis in the PLE dogs. The fold change relative gene expression of CD3e, CD4, CD5, CD8, and CD21 in PLE dogs was compared to the HC group using the Mann–Whitney *U* test. Statistical analyses were performed using GraphPad Prism (GraphPad Software Inc., La Jolla, California), and p-values ≤ 0.05 were considered significant.

3 | Results

3.1 | Dogs-Baseline Data

A total of 36 dogs, including 18 dogs with PLE and 18 HC dogs, were included in the study. The PLE cohort included nine females (all spayed) and nine males (all neutered) with a median age of 6 years (range 2–11 years) and a median weight of 26.5 kg

TABLE 1 | Baseline demographic characteristics and biochemical data in healthy control dogs and dogs with protein-losing enteropathy.

Variable	Reference interval	HC dogs median (range) or mean ± SD above[b] or below[c] RI	PLE dogs median (range) or mean ± SD above[b] or below[c] RI	p^a
Age (years)	NA	5 (3–11) NA	6 (2–11) NA	0.49
Body weight (kg)	NA	24 (3.6–42) NA	26 (3.1–46) NA	0.77
Lymphocyte ($\times 10^3/\mu L$)	0.8–3.9	2.4 ± 0.9	1.53 ± 0.8 1/18 (6%)[c]	0.004
Albumin (g/dL)	2.8–3.6	3.2 ± 0.2	1.6 ± 0.4 18/18 (100%)[c]	< 0.001
Globulin (g/dL)	2.3–3.7	3.0 ± 0.3	2.1 ± 0.4 13/18 (72%)[c]	< 0.001
Cholesterol (mG/dL)	126–325	271 ± 63 1/18 (6%)[b]	122 ± 63 7/18 (38%)[c]	< 0.001
Ionized calcium (mmol/L)	1.25–1.45	NA	1.34 (0.97–1.49) 3/18 (17%)[c]	NA
25OHD (nmol/L)	109–423	NA	46 (10–228) 14/18 (78%)[c]	NA
Cobalamin (ng/L)[d]	251–908	NA	175 (151–685) 11/18 (61%)[c]	NA

Abbreviations: 25OHD, 25-hydroxyvitamin-D; HC, healthy control; PLE, protein-losing enteropathy; RI, reference interval; SD, standard deviation.
[a]*p*-value as assessed by Mann–Whitney *U* for non-parametric variables (data presented as median (range)) and *t*-test for parametric variables (data presented as mean ± SD).
[b]Indicates proportion reported is the number of dogs above the RI.
[c]Indicates proportion reported is the number of dogs below the RI. If no proportions are reported, all dogs were within the RI.
[d]Concentrations reported as < 150 were recorded as 151 ng/L.

(range 3–46 kg). The control group consisted of nine females (all spayed) and nine males (all neutered) with a median age of 5 years (range 3–11 years) and a median weight of 23.8 kg (range 3.6–41.7 kg). Age and weight were not different between PLE and HC dogs (Table 1). Median BCS (0–9) [47] in dogs with PLE was 4 (range 2–5). Median BCS in the control group was 5 (range 4–7). Muscle condition score (MCS) [48] in dogs with PLE was normal for three dogs, mild for eight dogs, moderate for five dogs, and marked for two dogs. Muscle condition score was normal for all control dogs.

Three dogs in the PLE group were mixed breed. The remaining breeds were American Pit Bull ($n = 2$), Labrador Retriever ($n = 2$), Soft-coated Wheaten Terrier ($n = 2$), and one each of American Cocker Spaniel, American Mastiff, Bernese Mountain Dog, Coton de Tulear, Border Collie, Goldendoodle, Maltese/Yorkshire terrier, Papillon, and Rottweiler. Eight HC dogs were mixed breed. The remaining dogs included 1 each of the following—American Bulldog, Dalmatian, Goldendoodle, Golden retriever, Labradoodle, Labrador Retriever, Maltese/Yorkshire terrier, Pug, Tibetan Terrier, and Toy Poodle.

Reported clinical signs at the time of enrollment in the PLE dogs included diarrhea (17/18; 94%), weight loss (17/18; 94%), decreased appetite (10/18; 56%), vomiting (4/18; 22%), and clinical signs (e.g., abdominal distention, difficulty breathing) associated with cavity effusions (5/18; 28%). Median CCECAI score [41] was 9.5 (range, 5–18) and median Purina fecal score [46] at the time of initial presentation was 6/7 (range 4–7/7). All dogs in the HC group had a CCECAI score of < 3. Additional relevant biochemical data for both the PLE and control dogs are shown in Table 1.

Abdominal ultrasonography was performed in all PLE dogs. The most commonly reported abnormalities included peritoneal effusion (13/18; 72%), small intestinal hyperechoic mucosal striations (8/18; 44%), small intestinal wall thickening (10/18; 55%), mucosal speckling (4/18; 22%), and 3/18 (17%) dogs having mild abdominal lymphadenopathy.

Esophagogastroduodenoscopy was performed in 17/18 (94%) of dogs, and ileocolonoscopy was attempted in 16/17 (94%) and successful in 15/16 (94%). One dog had exploratory laparotomy at a first opinion clinic but remained naive to treatment prior to enrollment. All dogs had intestinal inflammatory infiltrates noted on intestinal biopsy. The severity of the infiltrate was noted to be moderate in 13/18 (72%) cases, marked in 3/18 (17%), and mild in 2/18 (11%) cases. The type of infiltrate was described as lymphoplasmacytic with eosinophils in 10/18 (56%) dogs and lymphoplasmacytic in 8/18 (44%) dogs. Mucosal infiltration of inflammatory cells involved the lamina propria in all 18 dogs, with additional epithelial involvement observed in 6/18 (33%) cases. Lacteal dilation was noted in 17/18 (94%) of cases, with the degree ranging from mild in 12/17 (71%) cases, moderate in

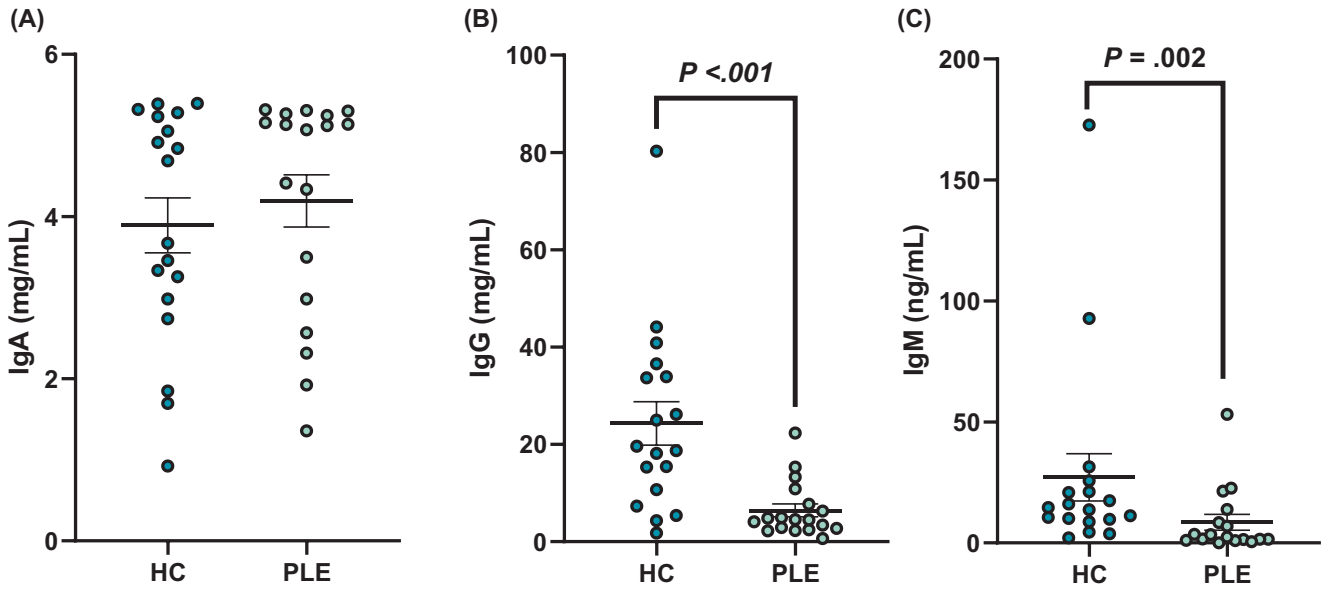

FIGURE 1 | (A) Boxplot showing serum concentrations of IgA in healthy control dogs and dogs with PLE. (B) Boxplot showing serum concentrations of IgG in healthy control dogs and dogs with PLE. (C) Boxplot showing serum concentrations of IgG in healthy control dogs and dogs with PLE. Median and interquartile range shown. HC, healthy control; PLE, protein-losing enteropathy.

3/17 (18%) cases, and marked in 2/17 (12%) cases. In nine cases, lacteal dilation was appreciated in deeper lymphatic vessels. Villous stunting occurred in 9/18 (50%) dogs and was predominantly mild (7/9; 78%), with moderate changes in the remaining 2/9 (22%) cases. Epithelial injury was similarly present in 9/18 (50%) dogs, characterized as mild in 7/9 (78%), moderate in 1/9 (11%), and marked in 1/9 (11%). Crypt dilation affected 10/18 (56%) dogs and was mild in all affected cases. All inflammatory infiltration and morphologic lesions of the small intestine were evaluated using a quantitative simplified scoring system [44].

3.2 | Serum Immunoglobulin Concentrations

Serum concentrations of immunoglobulins in the PLE dogs vs. HC are shown in Figure 1A–C. There was no difference in serum IgA concentrations between PLE (median 4.2 mg/mL, range 0.92–5.4 mg/mL) and HC (median 5.1 mg/mL, range 1.4–5.3 mg/mL) dogs ($p = 0.61$). Serum concentrations of IgG were lower in PLE dogs (median 4.5 mg/mL, range 0.67–22.4 mg/mL) when compared to HC dogs (median 19 mg/mL, range 1.8–80.3; $p < 0.001$). Dogs with PLE also had decreased concentrations of serum IgM (median 2.4 mg/mL, range 0.0009–53.1 mg/mL) when compared to HC dogs (median 14.2 mg/mL, range 2.1–172.8 mg/mL; $p = 0.002$). Serum immunoglobulin concentrations were not correlated with age or weight in either cohort (Figures S2–S5).

3.3 | Lymphocyte Evaluation

The total number of PMBCs (dead and viable) collected from both the HC and PLE cohorts is shown in Figure 2. Flow cytometry to characterize serum lymphocyte population was planned but was not able to be confidently performed as a majority of PLE dogs had too few cells available for analysis.

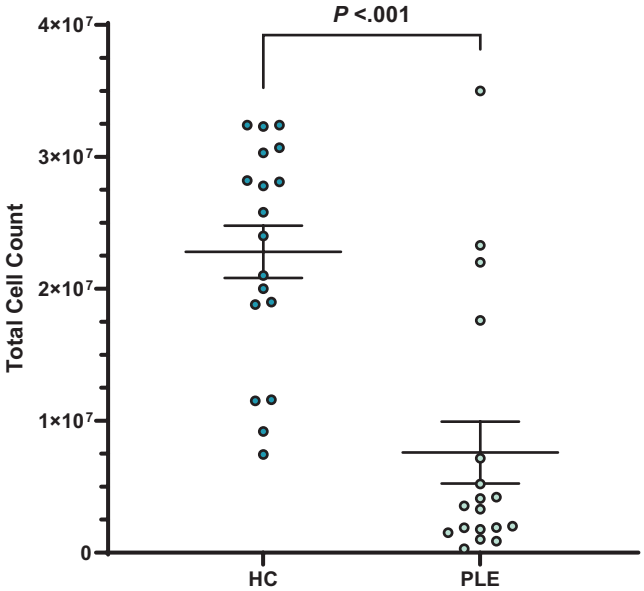

FIGURE 2 | Boxplot showing total peripheral blood mononuclear cells collected from healthy control dogs and dogs with PLE. Mean and standard deviation shown. HC, healthy control; PLE, protein-losing enteropathy.

Cell surface markers were evaluated by qPCR and revealed significant differences in gene expression in PLE dogs relative to HC dogs in CD3e ($p < 0.001$), CD5 ($p < 0.001$), and CD8 ($p = 0.007$) lymphocytes, while CD4 and CD21 gene expression was not different (Table 2 and Figure 3A–E).

4 | Discussion

Differences in serum immunoglobulin concentrations and gene expression of lymphocytes were observed in dogs with PLE

TABLE 2 | Relative gene expression (via quantitative PCR) of various immune cell markers in healthy control dogs and dogs with protein-losing enteropathy.

Relative gene expression[a]	HC dogs median (range)	PLE dogs median (range)	p[b]
CD3e[c]	0.92 (0.41–3.2)	0.24 (0.003–1.1)	< 0.001
CD4[d]	1 (0.25–5.6)	0.38 (0.01–10.4)	0.11
CD5[c]	0.94 (0.23–5)	0.17 (0.01–0.46)	< 0.001
CD8[d]	0.92 (0.32–2.4)	0.47 (0.06–1.7)	0.007
CD21[c]	1.5 (0.05–5)	0.27 (0.13–3.5)	0.06

Abbreviations: HC, healthy control; PLE, protein-losing enteropathy.
[a]Relative to the mean absolute expression in the healthy control group.
[b]p-value as assessed by Mann–Whitney U.
[c]Data available for $n = 14$ PLE dogs.
[d]Data available for $n = 15$ PLE dogs.

compared to a cohort of sex, age, and weight-matched healthy control dogs. Unlike in humans with PIL, differences in CD4 T-cell expression between PLE dogs and HC were not observed; rather, there was reduced expression of CD3e, CD5, and CD8 in PLE dogs relative to HC dogs. Thus, dogs with PLE have quantitative reductions in immune system components, similar to humans with IL.

Concentrations of serum immunoglobulins below the reference interval have been well documented in humans with both primary and secondary intestinal lymphangiectasia [17–19, 23, 30, 49–52]. Serum IgA, IgG, and IgM concentrations below the reference interval were reported in patients with PIL compared to healthy controls, with reductions most marked for serum IgG concentrations. Despite decreased concentrations, synthetic rates of immunoglobulin production remained normal, while fractional catabolic rates were increased. These findings indicate that decreased concentrations of immunoglobulins in these patients were likely due to excessive loss into the gastrointestinal tract, without a sufficient compensatory increase in production [19]. Multiple case series have reported similar findings of decreased concentrations of various serum immunoglobulins in humans with PIL [17, 18, 50, 52]. Children with PLE induced by the Fontan procedure have significantly reduced concentrations of serum IgG and IgA when compared to a group of children post-Fontan who did not develop PLE [23]. In contrast, humans with Crohn's disease and ulcerative colitis have normal to increased concentrations of serum immunoglobulins [53, 54]. Among 608 patients with Crohn's disease, elevated serum IgG, IgA, and IgM were found in 25%, 17%, and 2% of cases, respectively, with remaining patients having serum immunoglobulin concentrations within the reported reference intervals [54]. An additional investigation reported serum concentrations of IgG above the reference interval in 27 patients with ulcerative colitis and 21 patients with Crohn's disease when compared to healthy controls [53].

The dogs in our study exhibited histologic evidence of both chronic inflammatory enteritis and intestinal lymphangiectasia, findings consistent with previous reports of non-neoplastic chronic small intestinal disease causing PLE [1–3, 40, 55]. Though two dogs with marked IL were included, the majority of dogs had only mild lacteal dilation noted. Despite this, significantly decreased concentrations of both serum IgG and IgM concentrations were observed in PLE dogs compared to healthy controls, similar to humans with primary and secondary IL, and differing from humans with Crohn's disease and ulcerative colitis. This might suggest that the degree of intestinal lymphangiectasia could have been underestimated in these cases, considering it can be segmental and is considered challenging to diagnose [56, 57]. We can also consider that IL might be clinically important in this cohort of dogs despite being classified as histologically mild. Alternatively, increased intestinal permeability rather than IL might be responsible for immunoglobulin loss in some of these dogs with PLE [58]. Additionally, alterations in the normal production of immunoglobulins could explain why serum IgG and IgM concentrations were decreased in PLE dogs compared to healthy controls. Increased synthesis and secretion could explain why serum IgA concentrations in our PLE dogs were similar to HC. This would be supported by investigations in humans with IBD and the importance of IgA in mucosal immunity of the gastrointestinal tract [53, 54, 59]. Thus, it is possible that increases in the production of IgA in dogs with PLE might compensate for losses and thus concentrations are similar to HC dogs. Alternatively or additionally, IgA might have been preserved as less IgA is present in circulation when compared to IgG or IgM [60].

The original plan was to utilize flow cytometry to evaluate lymphocyte populations in our group of dogs with PLE compared to healthy controls. However, the laboratory advised that the number of PMBCs obtained from several PLE dogs was too few to draw accurate conclusions via flow cytometry. This did not appear to be related to PMBC storage time, as dogs with PLE collected at both the beginning and end of the study period were reported to have too few cells. While we had expected that absolute lymphocyte counts might be lower in dogs with PLE compared to HC, the degree of this difference was greater than anticipated, and thus required us to pivot to a different method of analysis for characterization of lymphocyte populations in PLE dogs. Quantitative PCR for gene expression allowed us to compare the abundance of gene transcripts between the two groups; however, we were unable to count individual cell populations as would have been possible with flow cytometry. Nonetheless, we did find that dogs with PLE had reduced expression of CD3e, CD5, and CD8 compared to the healthy control group, suggesting that the lymphocytes that bear these surface markers are less abundant within the total population of PMBCs in dogs with PLE compared to HC dogs. Similar to the reductions in serum concentrations of IgG and IgM, these results are similar to what is observed in humans with both primary and secondary IL [18–21, 23, 61, 62]. However, expression of CD4 was not different in dogs with PLE compared to HC. This differs from humans with IL where the CD4+ lymphocytes appear to be more selectively reduced compared to other lymphocyte populations [18, 20, 23, 61]. This selective reduction in CD4+ T cells is theorized to be related to trafficking of lymphocytes and evidence that certain populations of T-cells

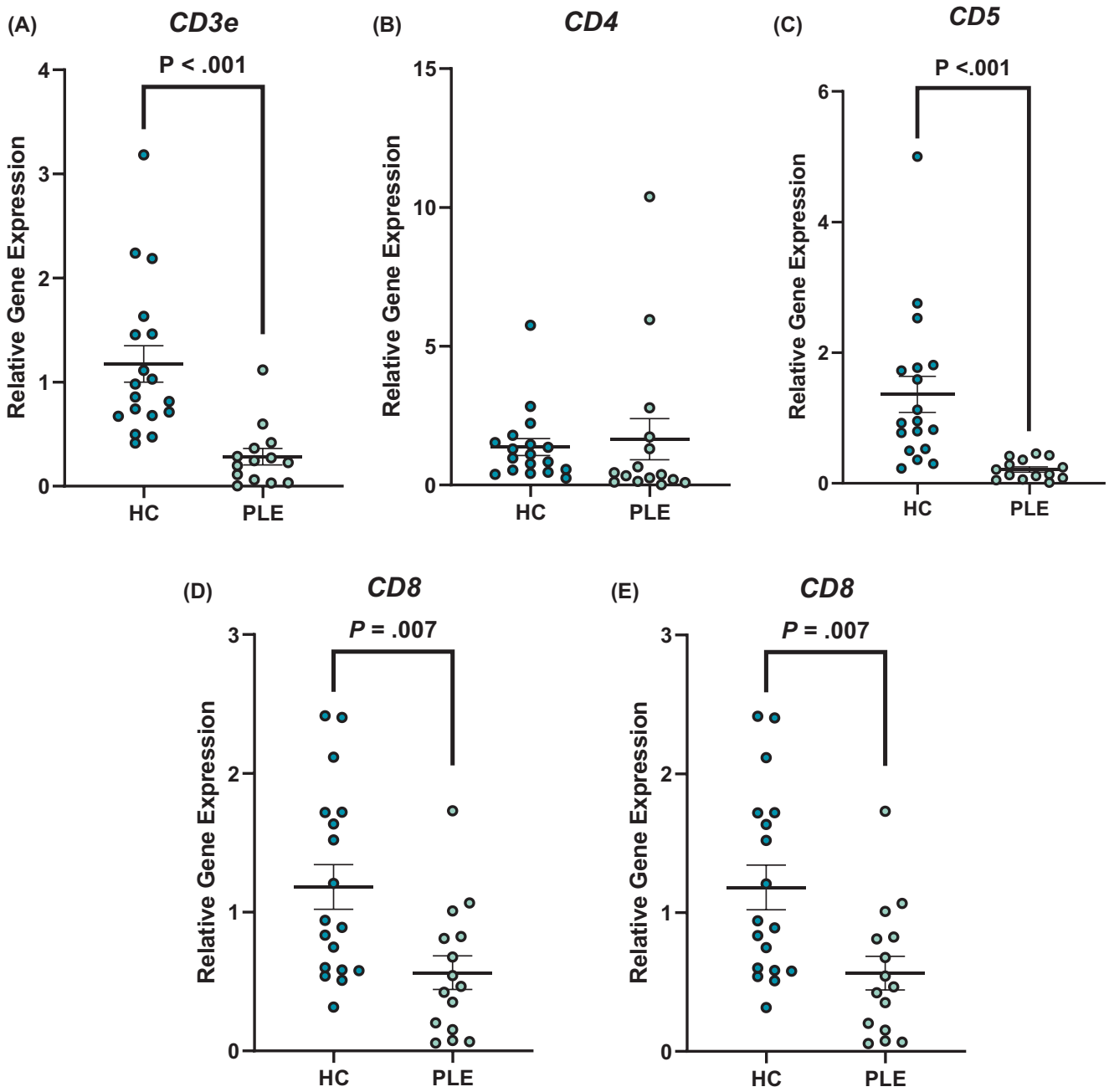

FIGURE 3 | (A) Boxplot depicting relative gene expression of CD3e in peripheral blood mononuclear cells of healthy control dogs and dogs with PLE. (B) Boxplot depicting relative gene expression of CD4 in peripheral blood mononuclear cells of healthy control dogs and dogs with PLE. (C) Boxplot depicting relative gene expression of CD5 in peripheral blood mononuclear cells of healthy control dogs and dogs with PLE. (D) Boxplot depicting relative gene expression of CD8 in peripheral blood mononuclear cells of healthy control dogs and dogs with PLE. (E) Boxplot depicting relative gene expression of CD21 in peripheral blood mononuclear cells of healthy control dogs and dogs with PLE. Median and interquartile range shown. HC, healthy control; PLE, protein-losing enteropathy.

are more likely to be recirculating through the lymphatics (and therefore more susceptible to loss) than in tissue [20, 23]. This theory is supported by the fact that naive CD45RA+ were the specific population most affected in a small group of humans with PIL [20]. While humans with IL have reduced populations of CD4+ T cells compared to healthy controls, humans with IBD have a wide variety of alterations to T cell subsets and functionality, reflecting the heterogeneity and complexity of the disorders that fall under the umbrella of IBD [63–67]. Additionally, dogs with CIE have higher concentrations of peripheral T helper lymphocytes and regulatory T cells when compared to healthy controls [68]. Since our cohort of dogs with PLE included dogs with various degrees of inflammatory infiltrates and intestinal lymphangiectasia, it is possible that the lack of difference in expression of CD4+ between our PLE dogs and HC reflects the heterogeneity in our cohort. In fact, the lack of difference might have been most influenced by 2 of the dogs with PLE who appeared to be outliers with increased expression of CD4 compared to the other dogs. The dog with the highest relative gene expression's histology showed only rare instances of mild lacteal

dilation while having a marked amount of lymphoplasmacytic and eosinophilic enteritis.

The implications of our findings of quantitative reductions in immune-system components in treatment-naïve PLE dogs are not yet clear. Despite the well-described immunodeficiency that occurs in humans with IL, the incidence of intestinal bacterial translocation and opportunistic infection appears to be low overall [19–21, 51], though infection might be more likely to occur in children with PIL [30, 49]. It is important to note, however, that humans with IL are not typically treated with immunosuppressive medications, so dogs with PLE with reductions in immune system components treated with immunosuppression could be at greater risk for infection. Our results suggest that dogs with PLE should be monitored closely for the development of infection. Our results might also support the suggestion for more conservative use of immunosuppressive medications in dogs with PLE. This aligns with recent literature demonstrating both the efficacy of dietary intervention [35–37, 39] and the absence of additional benefit when adding secondary immunosuppressive agents to glucocorticoid therapy in dogs with PLE [40]. Finally, our study suggests that dogs with PLE are more immunologically similar to humans with IL than to humans with IBD and seem to differ from a group of mostly normoalbuminemic dogs with CIE [68]. This might suggest that the IL could precede the inflammation in dogs with PLE or that the development of PLE syndrome in a dog with CIE might be attributed to the secondary IL rather than merely representing a more severe manifestation of the CIE spectrum. Alternatively, or in addition to these mechanisms, the immunologic abnormalities in dogs with PLE might result from increased intestinal permeability, as dogs with lymphoplasmacytic enteritis and hypoalbuminemia have increased intestinal permeability when compared to normoalbuminemic dogs with lymphoplasmacytic enteritis [58].

Because we were unable to perform flow cytometry for this study, we were not able to differentiate into specific subsets of lymphocytes, which would have allowed for a more precise understanding of the changes observed. The initial PMBC count was lower than desired prior to storage; thus, we would suggest a larger volume of blood might be required in the future to be able to perform accurate flow cytometric analysis of PMBCs in dogs with PLE. However, it is possible that alternative storage conditions could have also led to improved viability, allowing for adequate cell counts to perform the analysis. No qualitative analysis was performed, so whether the function of the immunoglobulins or lymphocytes is affected is unknown. Another limitation of our study is that the healthy control cohort was not breed matched to the PLE cohort. We initially attempted to breed match; however, it proved challenging to find healthy controls that met our inclusion criteria for rare breeds and for breeds with a predisposition for PLE. Finally, we considered evaluating for differences in immunoglobulins and expression of lymphocyte cell surface markers between subsets of our dogs with PLE (e.g., dogs with histologic evidence of mild vs. marked IL); however, the number of dogs in certain subsets was too few to perform such comparisons, and we feel that any such comparison would also be limited by our insensitive methods of diagnosing IL in dogs. Additionally, although no dogs required exclusion due to the weight requirement during the study period, the weight limitation could have excluded some dogs from breeds predisposed to PLE (e.g., Yorkshire Terriers).

In conclusion, dogs with treatment-naïve PLE have quantitative reductions in humoral and cell-mediated immune system components, similar to humans with both primary and secondary intestinal lymphangiectasia. These immunologic alterations could be considered in the development, therapeutic management, and monitoring of dogs with PLE.

Acknowledgments

The Attune CytPix, located in the Michigan State University Flow Cytometry Core Facility, is supported by the Equipment Grants Program, award #2022-70410-38419, from the U.S. Department of Agriculture (USDA), National Institute of Food and Agriculture (NIFA).

Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by Michigan State University Institutional Animal Care and Use Committee ID # PROTO202200456. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

References

1. K. Allenspach and C. Iennarella-Servantez, "Canine Protein Losing Enteropathies and Systemic Complications," *Veterinary Clinics of North America. Small Animal Practice* 51, no. 1 (2021): 111–122.

2. M. D. Craven and R. J. Washabau, "Comparative Pathophysiology and Management of Protein-Losing Enteropathy," *Journal of Veterinary Internal Medicine* 33, no. 2 (2019): 383–402.

3. S. A. Jablonski Wennogle, S. l. Priestnall, and C. b. Webb, "Histopathologic Characteristics of Intestinal Biopsy Samples From Dogs With Chronic Inflammatory Enteropathy With and Without Hypoalbuminemia," *Journal of Veterinary Internal Medicine* 31, no. 2 (2017): 371–376.

4. G. Rossi, M. Cerquetella, E. Antonelli, et al., "The Importance of Histologic Parameters of Lacteal Involvement in Cases of Canine Lymphoplasmacytic Enteritis," *Gastroenterology and Hepatology From Bed to Bench* 8, no. 1 (2015): 33–41.

5. T. A. Waldmann, J. L. Steinfeld, T. F. Dutcher, J. D. Davidson, and R. S. Gordon, "The Role of the Gastrointestinal System in "Idiopathic Hypoproteinemia"," *Gastroenterology* 41 (1961): 197–207.

6. T. A. Waldmann, S. Broder, and W. Strober, "Protein-Losing Enteropathies in Malignancy," *Annals of the New York Academy of Sciences* 230 (1974): 306–317.

7. P. Wilkinson, B. Pinto, and J. R. Senior, "Reversible Protein-Losing Enteropathy With Intestinal Lymphangiectasia Secondary to Chronic Constrictive Pericarditis," *New England Journal of Medicine* 273, no. 22 (1965): 1178–1181.

8. S. F. Pollack, A. L. Geffrey, E. A. Thiele, and U. Shah, "Primary Intestinal Lymphangiectasia Treated With Rapamycin in a Child With Tuberous Sclerosis Complex (TSC)," *American Journal of Medical Genetics. Part A* 167A, no. 9 (2015): 2209–2212.

9. J. Johansson, C. Sahin, R. Pestoff, et al., "A Novel SMAD4 Mutation Causing Severe Juvenile Polyposis Syndrome With Protein Losing Enteropathy, Immunodeficiency, and Hereditary Haemorrhagic Telangiectasia," *Case Reports in Gastrointestinal Medicine* 2015 (2015): 140616.

10. S. Bandikatla, S. Maharaj, A. Dadlani, K. Ramsubiek, and A. Rojan, "Thrombosis Secondary to Protein-Losing Enteropathy (PLE): A Case Report and Review," *Thrombosis Update* 3 (2021): 100050.

11. L. Zhang, D. K. W. Ocansey, L. Liu, et al., "Implications of Lymphatic Alterations in the Pathogenesis and Treatment of Inflammatory Bowel Disease," *Biomedicine & Pharmacotherapy* 140 (2021): 111752.

12. R. Hokari and A. Tomioka, "The Role of Lymphatics in Intestinal Inflammation," *Inflammation and Regeneration* 41, no. 1 (2021): 25.

13. F. L. Connor, S. Angelides, M. Gibson, et al., "Successful Resection of Localized Intestinal Lymphangiectasia Post-Fontan: Role of (99m) technetium-Dextran Scintigraphy," *Pediatrics* 112, no. 3 Pt 1 (2003): e242–e247.

14. R. H. Feldt, D. J. Driscoll, K. P. Offord, et al., "Protein-Losing Enteropathy After the Fontan Operation," *Journal of Thoracic and Cardiovascular Surgery* 112, no. 3 (1996): 672–680.

15. R. Hall, A. Nguyen, L. A. Panton, J. Shapiro, I. Gorbounova, and D. Hoytema Van Konijnenburg, "Primary Intestinal Lymphangiectasia as a Cause of Combined Immunodeficiency," *Journal of Human Immunity* 1, no. CIS2025 (2025): CIS2025abstract.19.

16. C. Müller, H. Wolf, J. Göttlicher, C. C. Zielinski, and M. M. Eibl, "Cellular Immunodeficiency in Protein-Losing Enteropathy. Predominant Reduction of CD3+ and CD4+ Lymphocytes," *Digestive Diseases and Sciences* 36, no. 1 (1991): 116–122.

17. H. Yamamoto, T. Tsutsui, M. Mayumi, and S. Kasakura, "Immunodeficiency Associated With Selective Loss of Helper/Inducer T Cells and Hypogammaglobulinaemia in a Child With Intestinal Lymphangiectasia," *Clinical and Experimental Immunology* 75, no. 2 (1989): 196–200.

18. D. Heresbach, J. L. Raoul, N. Genetet, et al., "Immunological Study in Primary Intestinal Lymphangiectasia," *Digestion* 55, no. 1 (1994): 59–64.

19. W. Strober, R. D. Wochner, P. P. Carbone, and T. A. Waldmann, "Intestinal Lymphangiectasia: A Protein-Losing Enteropathy With Hypogammaglobulinemia, Lymphocytopenia and Impaired Homograft Rejection," *Journal of Clinical Investigation* 46, no. 10 (1967): 1643–1656.

20. I. J. Fuss, W. Strober, B. A. Cuccherini, et al., "Intestinal Lymphangiectasia, a Disease Characterized by Selective Loss of Naive CD45RA+ Lymphocytes Into the Gastrointestinal Tract," *European Journal of Immunology* 28, no. 12 (1998): 4275–4285.

21. S. Sanges, N. Germain, S. Vignes, et al., "Protein-Losing Enteropathy as a Complication and/or Differential Diagnosis of Common Variable Immunodeficiency," *Journal of Clinical Immunology* 42, no. 7 (2022): 1461–1472.

22. Y. F. Cheung, H. Y. H. Tsang, and J. S. Y. Kwok, "Immunologic Profile of Patients With Protein-Losing Enteropathy Complicating Congenital Heart Disease," *Pediatric Cardiology* 23, no. 6 (2002): 587–593.

23. H. S. Magdo, T. L. Stillwell, M. J. Greenhawt, et al., "Immune Abnormalities in Fontan Protein-Losing Enteropathy: A Case-Control Study," *Journal of Pediatrics* 167, no. 2 (2015): 331–337.

24. X. Huber, L. Degen, S. Muenst, and M. Trendelenburg, "Primary Intestinal Lymphangiectasia in an Elderly Female Patient: A Case Report on a Rare Cause of Secondary Immunodeficiency," *Medicine* 96, no. 31 (2017): e7729.

25. S. Vignes and G. Carcelain, "Increased Surface Receptor Fas (CD95) Levels on CD4+ Lymphocytes in Patients With Primary Intestinal Lymphangiectasia," *Scandinavian Journal of Gastroenterology* 44, no. 2 (2009): 252–256.

26. D. Lenz, J. Hambsch, P. Schneider, et al., "Protein-Losing Enteropathy in Patients With Fontan Circulation: Is It Triggered by Infection?," *Critical Care* 7, no. 2 (2003): 185–190.

27. J. M. Cunningham, S. Nepal, and A. E. Truesdale, "Primary Intestinal Lymphangiectasia Diagnosed by Video Capsule Endoscopy in a Patient With Immunodeficiency Presenting With *Morganella morganii* Bacteraemia," *BML Case Reports* 13, no. 9 (2020): e235898.

28. F. Naranjo-Saltos, A. Hallo, C. Hallo, A. Mayancela, and A. Rojas, "Gastrointestinal Cryptococcosis Associated With Intestinal Lymphangiectasia," *Case Reports in Medicine* 2020 (2020): 7870154.

29. J. E. Na, J. E. Kim, S. Park, et al., "Experience of Primary Intestinal Lymphangiectasia in Adults: Twelve Case Series From a Tertiary Referral Hospital," *World Journal of Clinical Cases* 12, no. 4 (2024): 746–757.

30. M. P. Dierselhuis, J. J. Boelens, F. G. A. Versteegh, C. Weemaes, and N. M. Wulffraat, "Recurrent and Opportunistic Infections in Children With Primary Intestinal Lymphangiectasia," *Journal of Pediatric Gastroenterology and Nutrition* 44, no. 3 (2007): 382–385.

31. J. Wen, Q. Tang, J. Wu, Y. Wang, and W. Cai, "Primary Intestinal Lymphangiectasia: Four Case Reports and a Review of the Literature," *Digestive Diseases and Sciences* 55, no. 12 (2010): 3466–3472.

32. R. Huber, G. Semmler, A. Mayr, F. Offner, and C. Datz, "Primary Intestinal Lymphangiectasia in an Adult Patient: A Case Report and Review of Literature," *World Journal of Gastroenterology* 26, no. 48 (2020): 7707–7718.

33. M. J. A. M. Braamskamp, K. M. Dolman, and M. M. Tabbers, "Clinical Practice. Protein-Losing Enteropathy in Children," *European Journal of Pediatrics* 169, no. 10 (2010): 1179–1185.

34. R. N. Lopez and A. S. Day, "Primary Intestinal Lymphangiectasia in Children: A Review," *Journal of Paediatrics and Child Health* 56, no. 11 (2020): 1719–1723.

35. N. Nagata, H. Ohta, N. Yokoyama, et al., "Clinical Characteristics of Dogs With Food-Responsive Protein-Losing Enteropathy," *Journal of Veterinary Internal Medicine* 34, no. 2 (2020): 659–668.

36. A. J. Rudinsky, J. C. Rowe, and V. J. Parker, "Nutritional Management of Chronic Enteropathies in Dogs and Cats," *Journal of the American Veterinary Medical Association* 253, no. 5 (2018): 570–578.

37. S. A. Jablonski Wennogle, J. Stockman, and C. B. Webb, "Prospective Evaluation of a Change in Dietary Therapy in Dogs With Steroid-Resistant Protein-Losing Enteropathy," *Journal of Small Animal Practice* 62, no. 9 (2021): 756–764.

38. H. Okanishi, R. Yoshioka, Y. Kagawa, and T. Watari, "The Clinical Efficacy of Dietary Fat Restriction in Treatment of Dogs With Intestinal Lymphangiectasia," *Journal of Veterinary Internal Medicine* 28, no. 3 (2014): 809–817.

39. M. Myers, S. A. Martinez, J. T. Shiroma, A. T. Watson, and R. A. Hostutler, "Prospective Evaluation of Low-Fat Diet Monotherapy in Dogs With Presumptive Protein-Losing Enteropathy," *Journal of the American Animal Hospital Association* 59, no. 2 (2023): 74–84.

40. S. Salavati Schmitz, A. Gow, N. Bommer, L. Morrison, and R. Mellanby, "Diagnostic Features, Treatment, and Outcome of Dogs With Inflammatory Protein-Losing Enteropathy," *Journal of Veterinary Internal Medicine* 33, no. 5 (2019): 2005–2013.

41. K. Allenspach, B. Wieland, A. Gröne, and F. Gaschen, "Chronic Enteropathies in Dogs: Evaluation of Risk Factors for Negative Outcome," *Journal of Veterinary Internal Medicine* 21, no. 4 (2007): 700–708.

42. A. J. Gold, D. K. Langlois, and K. R. Refsal, "Evaluation of Basal Serum or Plasma Cortisol Concentrations for the Diagnosis of Hypoadrenocorticism in Dogs," *Journal of Veterinary Internal Medicine* 30, no. 6 (2016): 1798–1805.

43. C. Mansfield, "Practical Interpretation and Application of Exocrine Pancreatic Testing in Small Animals," *Veterinary Clinics of North America. Small Animal Practice* 43, no. 6 (2013): 1241–1260.

44. K. A. Allenspach, J. P. Mochel, Y. Du, et al., "Correlating Gastrointestinal Histopathologic Changes to Clinical Disease Activity in Dogs With Idiopathic Inflammatory Bowel Disease," *Veterinary Pathology* 56, no. 3 (2019): 435–443.

45. R. J. Washabau, M. J. Day, M. D. Willard, et al., "Endoscopic, Biopsy, and Histopathologic Guidelines for the Evaluation of Gastrointestinal Inflammation in Companion Animals," *Journal of Veterinary Internal Medicine* 24, no. 1 (2010): 10–26.

46. Purina Fecal Score Chart, 2024, https://www.proplanveterinarydiets.ca/wp-content/uploads/2016/04/PPPVD-Fecal-Scoring-Chart-EN-FINAL.pdf.

47. D. Laflamme, "Development and Validation of a Body Condition Score System for Dogs," *Canine Practice* 22, no. 4 (1997): 10–15.

48. L. M. Freeman, K. E. Michel, B. M. Zanghi, B. M. Vester Boler, and J. Fages, "Evaluation of the Use of Muscle Condition Score and Ultrasonographic Measurements for Assessment of Muscle Mass in Dogs," *American Journal of Veterinary Research* 80, no. 6 (2019): 595–600.

49. S. B. Ingle and C. R. Hinge Ingle, "Primary Intestinal Lymphangiectasia: Minireview," *World Journal of Clinical Cases* 2, no. 10 (2014): 528–533.

50. J. Lynn, A. K. Knight, M. Kamoun, and A. I. Levinson, "A 55-Year-Old Man With Hypogammaglobulinemia, Lymphopenia, and Unrelenting Cutaneous Warts," *Journal of Allergy and Clinical Immunology* 114, no. 2 (2004): 409–414.

51. S. Vignes and J. Bellanger, "Primary Intestinal Lymphangiectasia (Waldmann's Disease)," *Orphanet Journal of Rare Diseases* 3 (2008): 5.

52. R. U. Sorensen, T. C. Halpin, C. R. Abramowsky, et al., "Intestinal Lymphangiectasia and Thymic Hypoplasia," *Clinical and Experimental Immunology* 59, no. 1 (1985): 217–226.

53. R. P. MacDermott, G. S. Nash, I. O. Auer, et al., "Alterations in Serum Immunoglobulin G Subclasses in Patients With Ulcerative Colitis and Crohn's Disease," *Gastroenterology* 96, no. 3 (1989): 764–768.

54. D. J. Song, J. Shen, M. H. Chen, et al., "Association of Serum Immunoglobulins Levels With Specific Disease Phenotypes of Crohn's Disease: A Multicenter Analysis in China," *Frontiers in Medicine* 8 (2021): 621337.

55. O. Dossin and R. Lavoué, "Protein-Losing Enteropathies in Dogs," *Veterinary Clinics of North America. Small Animal Practice* 41, no. 2 (2011): 399–418.

56. R. N. Larson, J. A. Ginn, C. M. Bell, M. J. Davis, and D. S. Foy, "Duodenal Endoscopic Findings and Histopathologic Confirmation of Intestinal Lymphangiectasia in Dogs," *Journal of Veterinary Internal Medicine* 26, no. 5 (2012): 1087–1092.

57. S. A. Jablonski Wennogle, S. L. Priestnall, A. Suárez-Bonnet, S. Soontararak, and C. B. Webb, "Lymphatic Endothelial Cell Immunohistochemical Markers for Evaluation of the Intestinal Lymphatic Vasculature in Dogs With Chronic Inflammatory Enteropathy," *Journal of Veterinary Internal Medicine* 33, no. 4 (2019): 1669–1676.

58. S. Kobayashi, K. Ohno, K. Uetsuka, et al., "Measurement of Intestinal Mucosal Permeability in Dogs With Lymphocytic-Plasmacytic Enteritis," *Journal of Veterinary Medical Science* 69, no. 7 (2007): 745–749.

59. R. P. MacDermott, G. S. Nash, M. J. Bertovich, M. V. Seiden, M. J. Bragdon, and M. G. Beale, "Alterations of IgM, IgG, and IgA Synthesis and Secretion by Peripheral Blood and Intestinal Mononuclear Cells From Patients With Ulcerative Colitis and Crohn's Disease," *Gastroenterology* 81, no. 5 (1981): 844–852.

60. A. J. German, E. J. Hall, and M. J. Day, "Measurement of IgG, IgM and IgA Concentrations in Canine Serum, Saliva, Tears and Bile," *Veterinary Immunology and Immunopathology* 64, no. 2 (1998): 107–121.

61. A. Koch, M. Hofbeck, H. Feistel, G. Buheitel, and H. Singer, "Circumscribed Intestinal Protein Loss With Deficiency in CD4+ Lymphocytes After the Fontan Procedure," *European Journal of Pediatrics* 158, no. 10 (1999): 847–850.

62. J. Moosmann, O. Toka, S. Lukassen, et al., "Lymphocyte Immune Response and T Cell Differentiation in Fontan Patients With Protein-Losing Enteropathy," *Thoracic and Cardiovascular Surgeon* 69, no. S 03 (2021): e10–e20.

63. R. Kalla, A. T. Adams, N. T. Ventham, et al., "Whole Blood Profiling of T-Cell-Derived microRNA Allows the Development of Prognostic Models in Inflammatory Bowel Disease," *Journal of Crohn's & Colitis* 14, no. 12 (2020): 1724–1733.

64. A. M. Globig, N. Hennecke, B. Martin, et al., "Comprehensive Intestinal T Helper Cell Profiling Reveals Specific Accumulation of IFN-γ+IL-17+Coproducing CD4+ T Cells in Active Inflammatory Bowel Disease," *Inflammatory Bowel Diseases* 20, no. 12 (2014): 2321–2329.

65. R. Gomez-Bris, A. Saez, B. Herrero-Fernandez, C. Rius, H. Sanchez-Martinez, and J. M. Gonzalez-Granado, "CD4 T-Cell Subsets and the Pathophysiology of Inflammatory Bowel Disease," *International Journal of Molecular Sciences* 24, no. 3 (2023): 2696.

66. C. Smids, C. S. Horjus Talabur Horje, J. Drylewicz, et al., "Intestinal T Cell Profiling in Inflammatory Bowel Disease: Linking T Cell Subsets to Disease Activity and Disease Course," *Journal of Crohn's & Colitis* 12, no. 4 (2018): 465–475.

67. J. Maul, C. Loddenkemper, P. Mundt, et al., "Peripheral and Intestinal Regulatory CD4+ CD25(High) T Cells in Inflammatory Bowel Disease," *Gastroenterology* 128, no. 7 (2005): 1868–1878.

68. B. Agulla, A. Villaescusa, Á. Sainz, et al., "Peripheral and Intestinal T Lymphocyte Subsets in Dogs With Chronic Inflammatory Enteropathy," *Journal of Veterinary Internal Medicine* 38, no. 3 (2024): 1437–1448.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.